Huawei Access Network SOP for Resolving QoS/Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on HEB-ACC-75 at Hebbal Edge

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_6063

Description

This Standard Operating Procedure (SOP) is designed to address a Major alarm (ALARM_6063) indicative of Quality of Service (QoS) / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Access node (HEB-ACC-75) at Hebbal Edge, operating under Huawei's CLI dialect. The failure domain is a critical concern, as it impacts the network's ability to ensure end-to-end service level agreements (SLAs) for users. The underlying cause of this issue stems from the interaction between the Access node's hardware and software components, specifically the QoS configuration and policing mechanisms. Huawei's Access node employs a hierarchical QoS architecture, where the QoS policy is applied at the ingress interface, classifying traffic into various classes, and prioritizing them based on their 802.1p CoS values. However, when the configured QoS policy is not effectively enforced due to microbursts, interface tail drops, or QoS queue starvation, it leads to a degradation in network performance, resulting in the identified alarm.

Pre-requisite Checks

- check: Verify the redundant link status on HEB-ACC-75 at Hebbal Edge to ensure it is up and functioning correctly.

- check: Dispatch remote hands to Hebbal Edge site to facilitate any necessary physical interventions.

- check: Review the current QoS configuration on HEB-ACC-75 to identify any potential issues or misconfigurations.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy on HEB-ACC-75	High
Inadequate policing mechanisms on the Access node	Medium
Hardware-related issues with the Access node's interface	Low

Diagnostic Commands & Expected Outputs

```
Command: display queuing statistics interface GigabitEthernet0/0/0/0, Expected Output: Queue Statistics for GigabitEthernet0/0/0/0: Queue ID 0: Drop Count 10, Packet Count 1000 Queue ID 1: Drop Count 20, Packet Count 2000, Description: This command displays the queue statistics
```

```
for the specified interface, helping to identify any potential issues with QoS queue starvation.

Command: display interface GigabitEthernet0/0/0/0 statistics, Expected Output: Interface GigabitEthernet0/0/0/0: Input Packets 100000, Output Packets 100000 Input Drops 10, Output Drops 20, Description: This command displays the interface statistics, including input and output packets, drops, and other relevant information to diagnose interface-related issues.

Command: display qos policy interface GigabitEthernet0/0/0/0, Expected Output: QoS Policy for GigabitEthernet0/0/0/0: Class Map Class-Default: Police 1000000 bps Police 1000000 bps, Description: This command displays the QoS policy applied to the specified interface, allowing for the verification of policing mechanisms and QoS configuration.
```

Resolution Steps

Step 1: Step: Verify and correct the QoS policy configuration on HEB-ACC-75, Command: undo qos policy interface GigabitEthernet0/0/0/0, Description: This command removes the existing QoS policy from the interface, allowing for the verification and correction of the configuration.

Step 2: Step: Configure the policing mechanisms on the Access node, Command: qos policy interface GigabitEthernet0/0/0/0, Description: This command configures the policing mechanisms on the Access node, ensuring that the QoS policy is enforced effectively.

Step 3: Step: Verify the QoS configuration and policing mechanisms, Command: display qos policy interface GigabitEthernet0/0/0/0, Description: This command verifies the QoS configuration and policing mechanisms, ensuring that they are correctly applied to the interface.

Post-Check Verification

```
Command: display queuing statistics interface GigabitEthernet0/0/0/0, Expected Output: Queue Statistics for GigabitEthernet0/0/0/0: Queue ID 0: Drop Count 0, Packet Count 10000 Queue ID 1: Drop Count 0, Packet Count 20000, Description: This command verifies that the queue statistics indicate no drops or issues with QoS queue starvation.

Command: display interface GigabitEthernet0/0/0/0 statistics, Expected Output: Interface GigabitEthernet0/0/0/0: Input Packets 1000000, Output Packets 1000000 Input Drops 0, Output Drops 0, Description: This command verifies that the interface statistics indicate no issues with interface-related drops or congestion.

Command: display qos policy interface GigabitEthernet0/0/0/0, Expected Output: QoS Policy for GigabitEthernet0/0/0/0: Class Map Class-Default: Police 1000000 bps Police 1000000 bps, Description: This command verifies that the QoS policy is correctly applied and enforced on the interface.
```

Roll-back Procedures

- Step: Reapply the previous QoS policy configuration, Command: qos policy interface GigabitEthernet0/0/0/0, Description: This command reverts the QoS policy configuration to its previous state, in case the fix fails.

- Step: Restore the previous policing mechanisms, Command: undo qos policy interface GigabitEthernet0/0/0/0, Description: This command removes the new policing mechanisms and restores the previous configuration, in case the fix fails.